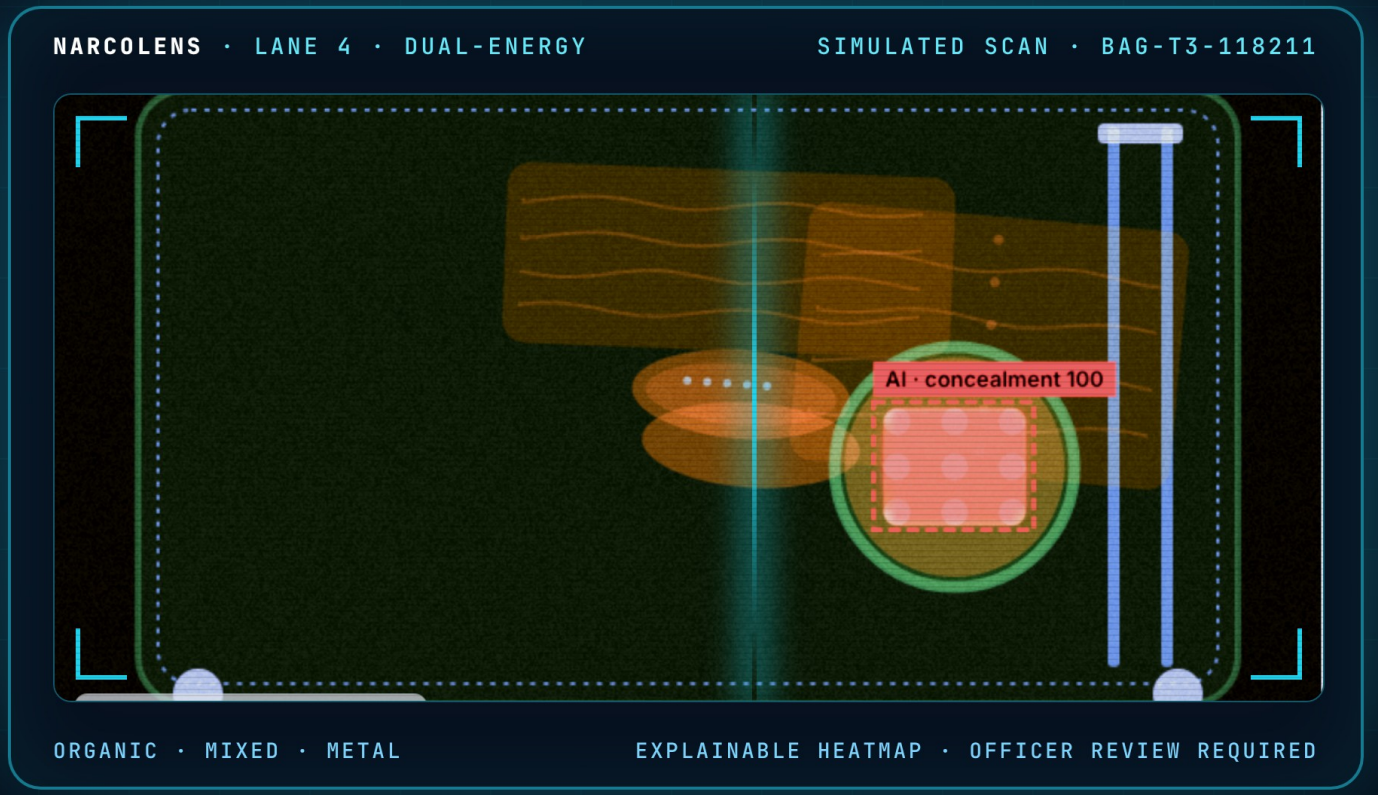
PRESIGHT EUREKA 3.0 · DEFENSE & POLICING

NarcoLens

From Scan to Evidence.

AI-Assisted Substance Screening for Police & Airport Customs

Transforming existing customs scanners into an intelligent, explainable narcotics screening network.



Simulated scan. Prototype output on synthetic data.

Farah Shaik & Riazat Rasool

Drug concealment is evolving faster than traditional inspection workflows

502
seizures

406 kg
narcotic and psychotropic substances

2.3 M+
tablets

Dubai Customs, January to May 2026. Source: Emirates News Agency (WAM), June 2026; also reported by Gulf News and Khaleej Times.

Concealment keeps changing

- Drugs inside clothing or buttons
- Narcotics in food or protein containers
- Tablets in legitimate packaging
- Synthetic substances that keep changing
- Complex parcel and cargo concealment

The challenge is no longer simply scanning more items. **It is understanding suspicious items faster and more intelligently.**

Today's scanner can show an anomaly. The officer must still determine what it means.

1 Scanner → 2 Operator interpretation → 3 Manual inspection → 4 Possible field test → 5 Laboratory confirmation → 6 Case documentation

- Heavy dependence on operator experience
- Suspicious regions can be hard to interpret
- Substance identity often needs additional testing
- Supporting evidence sits in different systems
- Emerging synthetic drugs keep creating new detection challenges
- What one site learns may not automatically benefit another

Detection exists. Intelligence remains fragmented.

NarcoLens adds an intelligence layer to the equipment customs already owns

EXPLAINABLE SUBSTANCE SCREENING

Officer review and action

NarcoLens intelligence layer

Software and edge AI running beside existing infrastructure

Existing scanners · desk camera · laboratory referral

NarcoLens

- Detects
- Correlates
- Assesses
- Explains
- Records

The officer

- Reviews
- Re-scans
- Opens
- Clears
- Escalates
- Seizes

AI recommends. Officers decide.

One digital case for every suspicious bag or parcel, with evidence added by multiple AI modules.

Five AI lenses. One assessment.

M1

X-Ray Vision

AI segmentation identifies suspicious organic concealment regions in luggage or cargo.

Output: AI heatmap and concealment probability

M2

Pills, Vials & Labels

Computer vision analyses tablet shape, size, colour, markings and packaging.

Output: Tablet and package similarity assessment

M3

Hyperspectral ID

Short-wave infrared spectra are compared with a reference substance library.

Output: Probable substance identification

NarcoLens Risk Engine

One item.
Five intelligence layers.
One explainable assessment.

M4

Novel Drug Predictor

Graph-based AI is proposed to predict signatures of emerging synthetic variants for forensic validation.

Output: Candidate spectral signatures

M5

Explainable Risk Fusion

Combines the available evidence into one risk score and shows each signal's contribution.

Output: Explainable case risk

Explainable assessment

One fused score with the contribution of every signal. The officer reviews and acts.

From first scan to laboratory referral

01

Scan

A bag, cargo item or parcel enters an existing scanner.

02

Detect

NarcoLens highlights suspicious concealment regions.

03

Inspect

The officer opens or photographs the object when required.

04

Identify

Tablet analysis and hyperspectral screening add evidence.

05

Fuse

Evidence is combined into an explainable risk score.

06

Act

The officer chooses the action. Seizures need supervisor approval.

CLEAR | RE-SCAN | OPEN | SEIZE

07

Record

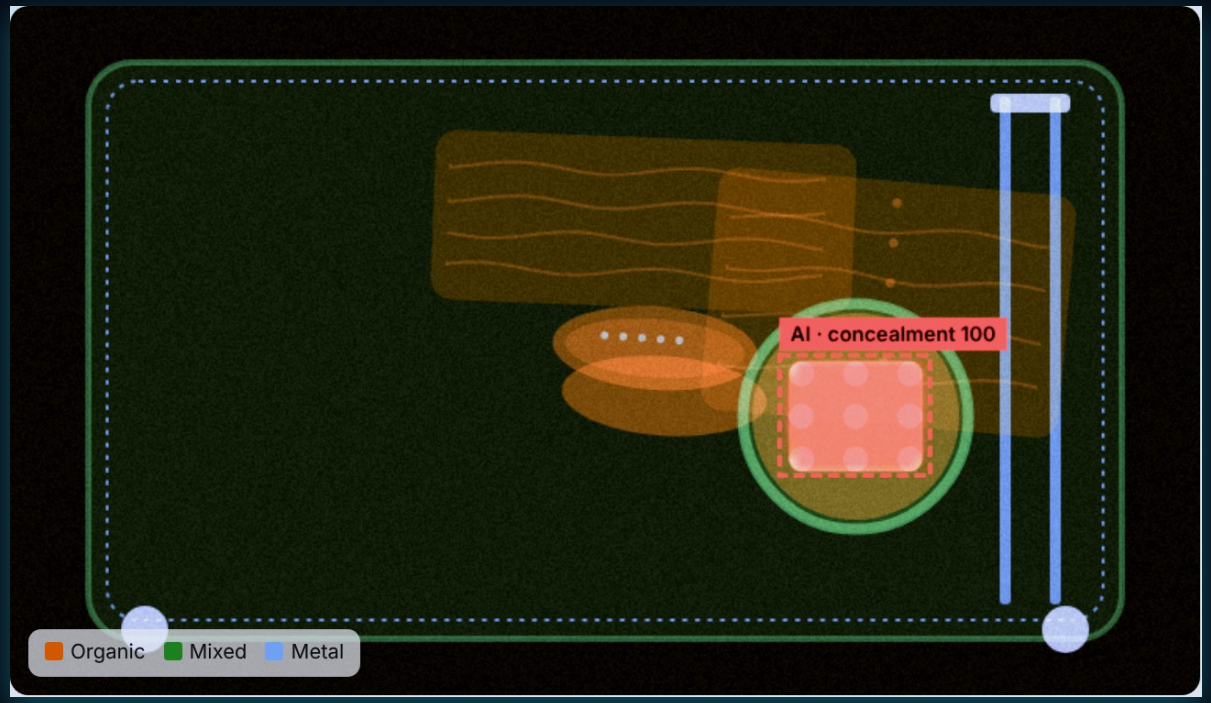
Every action joins one timestamped evidence trail.

08

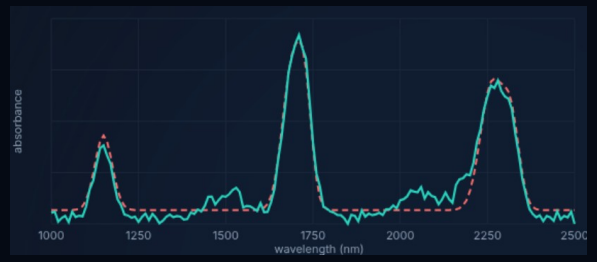
Confirm

Laboratory results can confirm the substance and feed validated learning.

Every alert comes with its evidence



Illustrative case. Simulated scan with the AI heatmap.



Spectral match against the library

Officer timeline

1 Flagged by M1
2 Bag opened
3 Spectrum attached
4 Seized, lab referral

Fused risk score

92
RISK

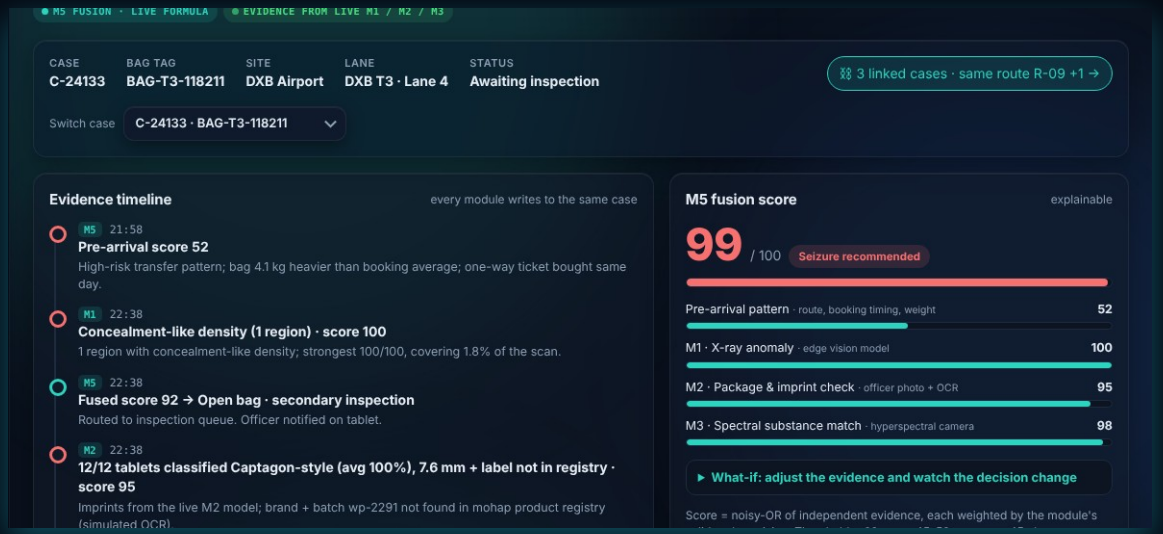
Illustrative example. Each bar shows one signal. The score is a weighted noisy-OR over the available evidence.

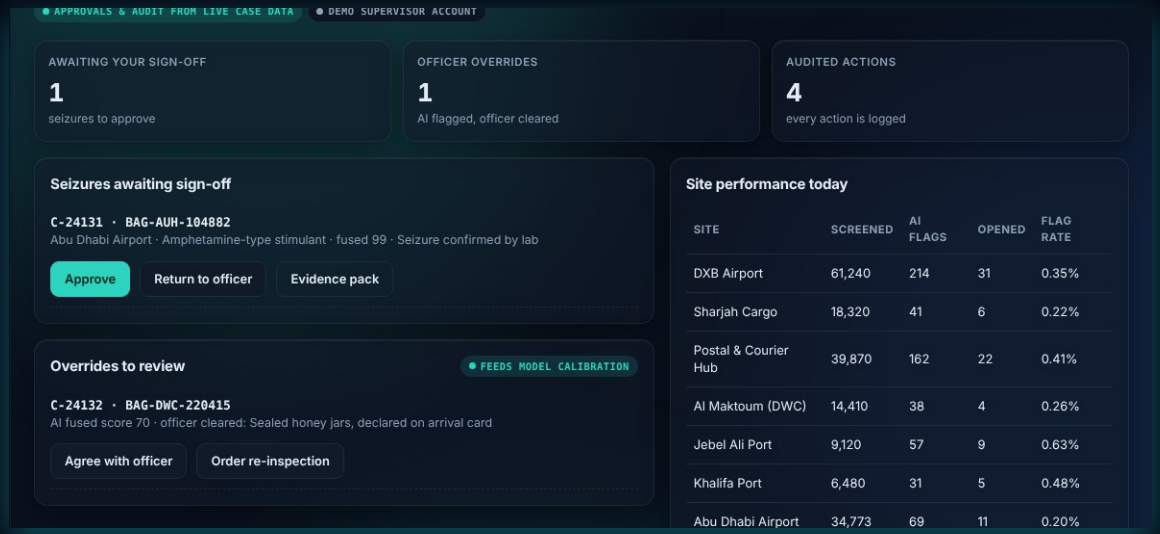
Pre-arrival risk	0.20
X-ray concealment	0.75
Tablet / package analysis	0.50
Hyperspectral match	0.55

Evidence, not just an alert

- Faster officer review
- Findings that stand up to review
- Stronger evidence continuity
- Easier supervisor review and laboratory referral

Two roles. One accountable workflow.





Inspector mode

Frontline screening and evidence

- Screens bags on the live lane and reviews the AI heatmap
- Checks tablets and labels and runs hyperspectral scans
- Reviews the fused score and the evidence timeline
- Asks NarcoLens why an item was flagged
- Clears, re-scans or opens the item and refers a seizure for sign-off

Supervisor mode

Oversight and accountability

- Approves seizures once the laboratory confirms the substance
- Reviews cases where the AI flagged and the officer cleared
- Compares screening and flag rates across entry points
- Reads the audit log of officer, laboratory and supervisor actions

Every action by either role is time-stamped in one audit trail. Demo accounts in the prototype are fictional.

A seizure at one entry point can strengthen awareness across the network



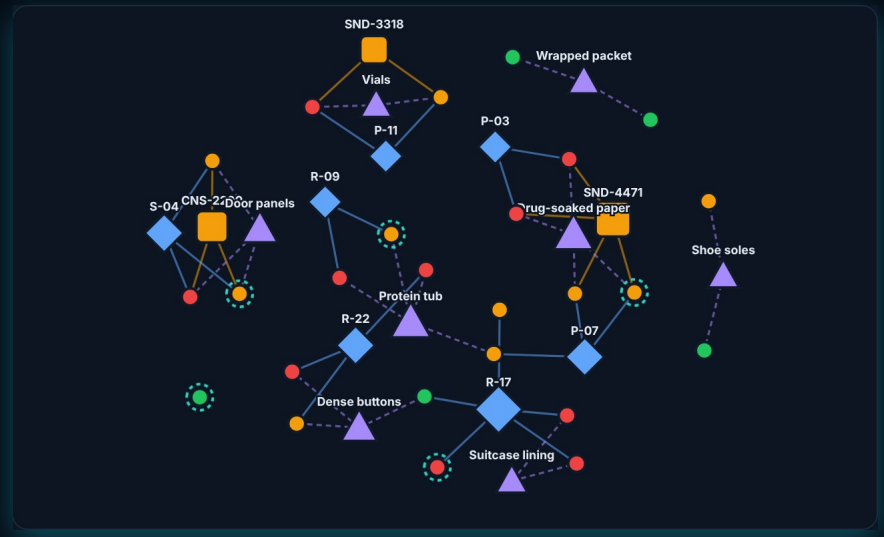
Seven simulated entry points

Airports | Seaports | Postal hubs | Courier facilities | Cargo terminals

Three suspicious parcels. Two entry points. One sender pattern.

Network analytics link cases by

- Sender
- Route
- Concealment method
- Packaging pattern
- Related case characteristics



Case link graph on simulated cases

The intelligence can travel without the raw inspection image

AT EACH SITE

SOVEREIGN CORE

1
Local inspection data
Stays at the site

2
Local model improvement
Trained on site

3
Protected model update
Encrypted and noised

4
Sovereign aggregation
Abnormal updates rejected

5
Updated model returned
To participating sites

PROPOSED · SIMULATED IN THE PROTOTYPE

- Raw inspection images remain at the site
- Protected model updates are shared instead
- Updates can be encrypted
- Differential privacy techniques are proposed
- Robust aggregation can help reject abnormal or poisoned updates
- Sovereign government infrastructure can host the central services



Federated learning simulator in the prototype

Learn together. Keep sensitive inspection data local.

From concept to working proof of concept

LIVE IN THE PROTOTYPE

- X-ray concealment AI
- Tablet and imprint classifier
- Hyperspectral matching algorithm
- Explainable risk fusion
- Offline web interface

SIMULATED OR PROPOSED

- Novel-drug prediction workflow
- Federated learning cycle
- Multi-site operational data

M1 X-RAY MODEL

100%

Bag-level recall

311 / 311 synthetic concealment bags

1.7%

False alarms

on synthetic clean bags

99.2%

Bag-level accuracy

595 / 600 synthetic bags

M2 TABLET MODEL

99.8%

Four-class accuracy

1,200 synthetic tablet images

100%

Twin-crescent recall

synthetic test set

Held-out test sets

600 synthetic bags
1,200 synthetic tablet images
Never seen in training

Results are from held-out synthetic test data and are not estimates of real-world field performance.

Current Technology Readiness: TRL 3

TRL 1 Basic principles

TRL 2 Technology concept

TRL 3 **Experimental proof of concept** — NARCOLENS TODAY

TRL 4 Lab validation

TRL 5 Relevant environment

TRL 6 Demonstration

TRL 7 Operational prototype

TRL 8 Qualified system

TRL 9 Proven operational deployment

NarcoLens currently demonstrates

- End-to-end workflow
- Working AI components
- Synthetic test environment
- Explainable case management
- Multi-module integration

Next major milestone

Validation using real customs scanner data

Start invisible. Measure everything.

Shadow mode: NarcoLens operates beside existing customs procedures and takes no operational action.

PHASE 1

Data preparation

Controlled integration with existing scanner outputs

PHASE 2

Shadow-mode operation

Runs on real scanner traffic with no operational effect

PHASE 3

Measure

Performance and officer workflow against real outcomes

PHASE 4

Calibrate and retrain

Risk fusion tuned and models retrained on validated real-world data

PHASE 5

Controlled deployment

Operational use after agreed targets are met

PHASE 3 MEASURES

- Detection recall
- False-alarm rate
- Scanner-vendor performance
- Officer review time
- Assessment time
- Evidence-record completeness

Proposed pilot acceptance target

≤ 5%

false alarms per lane

A proposed acceptance criterion, not an achieved real-world result.

Why NarcoLens is different

Capability	Traditional screening	NarcoLens
X-ray scanning	✓	✓
AI concealment localisation	Limited / system dependent	✓
Tablet visual verification	Separate process	✓
Hyperspectral identification	Separate tool	✓
Multi-sensor risk fusion	Fragmented	✓
Explainable AI	Limited	✓
Cross-site intelligence	Fragmented	✓ **on simulated cases**
Privacy-preserving shared learning	Limited	**Proposed** ✓
Unified case evidence trail	Multiple systems	✓

Traditional screening describes typical current workflows; capabilities vary by site and scanner vendor. NarcoLens items are demonstrated in the prototype on simulated data, except those marked proposed.

NarcoLens does not add another scanner. **It connects the intelligence around the scanner.**

From reactive inspection to intelligent interdiction

FASTER

Reduce the time required to understand a suspicious item.

SMARTER

Combine multiple signals instead of relying on a single detection method.

CONNECTED

Identify patterns across parcels, routes and entry points.

DEFENSIBLE

Maintain an explainable evidence trail from first scan to laboratory referral.

POTENTIAL OPERATIONAL OUTCOMES TO VALIDATE

- Fewer unnecessary manual inspections
- Faster assessment of suspicious items
- Better prioritisation of inspections
- Stronger laboratory referral records
- Improved intelligence across sites
- Faster adaptation to emerging synthetic drugs

Hypotheses to test in the pilot. They are not achieved results.

Every scan becomes intelligence

Today, customs systems inspect millions of items independently.

NarcoLens envisions a future where every validated seizure can improve the intelligence available to the next officer, at the next scanner, at the next border entry point.



UAE entry points (simulated network)

Detect earlier.
Understand faster.
Learn together.

Help us take NarcoLens from simulation to the border

ACCESS

Anonymised or controlled real scanner data for validation.

EXPERTISE

Collaboration with customs officers, forensic laboratories and anti-narcotics specialists.

PILOT

A controlled shadow-mode deployment at a selected customs environment.

NarcoLens

From Scan to Evidence.

AI intelligence for the officers protecting our borders.

Farah Shaik & Riazat Rasool